

EVALUATING MEMORY STRUCTURE IN LLM AGENTS

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ABSTRACT

Modern LLM-based agents and chat assistants rely on long-term memory frameworks to store reusable knowledge, recall user preferences, and augment reasoning. As researchers create more complex memory architectures, it becomes increasingly difficult to analyze their capabilities and guide future memory designs. Most long-term memory benchmarks focus on simple fact retention, multi-hop recall, and time-based changes. While undoubtedly important, these capabilities can often be achieved with simple retrieval-augmented LLMs and do not test complex memory hierarchies. To bridge this gap, we propose StructMemEval — a benchmark that tests the agent’s ability to organize its long-term memory, not just factual recall. We gather a suite of tasks that humans solve by organizing their knowledge in a specific structure: transaction ledgers, to-do lists, trees and others. Our initial experiments show that simple retrieval-augmented LLMs struggle with these tasks, whereas memory agents can reliably solve them if prompted how to organize their memory. However, we also find that modern LLMs do not always recognize the memory structure when not prompted to do so. This highlights an important direction for future improvements in both LLM training and memory frameworks.

1 INTRODUCTION

As Large Language Models (LLMs) keep improving, we trust them with increasingly longer and more difficult tasks (Kwa et al., 2025). Frontier models assist users over long stretches of time (Xu et al., 2022; Jang et al., 2023; Wang et al., 2025a), work with large documents (Packer et al., 2023; Chevalier et al., 2023), perform long-form reasoning (ARC Prize Foundation, 2024; Phan et al., 2025) and agentic tasks (Erdogan et al., 2025; Merrill et al., 2026). To solve these tasks, a model needs to process vast amounts of information, both input sources and its own workings. As LLMs deal with harder problems, it becomes infeasible to keep all that information in their working memory.

Recent works circumvent this by equipping the LLM with long-term memory: allowing the agent to offload its knowledge from working memory to an external database and retrieve it only when necessary (Packer et al., 2023; OpenAI, 2024). Memory designs vary between methods: from simple look-up dictionaries (Packer et al., 2023; Wang et al., 2024b) to complex frameworks (Chhikara et al., 2025; Rasmussen et al., 2025) and entire operating systems (Li et al., 2025; Kang et al., 2025).

To compare between memory architectures, researchers and practitioners evaluate them on specialized benchmarks like LOCOMO (Maharana et al., 2024) and LongMemEval (Wu et al., 2024). These benchmarks typically let the LLM process a scenario (e.g. personal assistant, corporate, agent), then ask questions to test its recall: single-hop, multi-hop, time-sensitive updates and more. However, recent works have found that simple memory recall often does not require complex memory architecture (Zhou & Han, 2025; Ai et al., 2025). In one such work, Zhou & Han (2025) demonstrates that simple retrieval can outperform complex memory hierarchies on both LOCOMO and LongMemEval. This leads to a question: what kinds of problems *do* require complex memory hierarchies?

In this work, we focus on one answer: that *memory allows LLM agents to organize their knowledge*. Instead of retrieving messages as-is, memory-augmented LLMs can extract knowledge and assemble it into a structure that best fits their task. We analyze scenarios that require not just factual recall (where retrieval is enough), but high-level memory organization. To that end, we gather a suite of tasks that a “notepad-augmented human” would solve by maintaining ledgers, drawing trees, tracking states and similar. We find that even simple tasks of this kind prove difficult for retrieval-augmented agents, but proper memory-augmented agents fare significantly better. When we explicitly prompt LLM agents to organize their memory, they can reliably solve tasks on a scale that retrieval-augmented LLMs struggle with, but the same agents perform less reliably without such “hints”. Informally, LLMs that can solve abstract algorithmic tasks often fail to recognize the same structure patterns in the wild. We release our task suite at <https://github.com/yandex-research/StructMemEval>.

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2 BACKGROUND

Transformer Working Memory. Transformer LLMs normally store task-specific information in the form of a Key-Value (KV) cache. This cache contains token-level vector representations used by attention layers on every inference step (Vaswani et al., 2017). For modern LLMs, this cache is limited to few $\times 10^4$ – 10^5 tokens with a few extreme cases going over a million tokens (GLM et al., 2024; Yang et al., 2025). While this limit can be extended (Peng et al., 2024; Li et al., 2023; Pekelis et al., 2024), a large KV cache fills accelerator memory¹ and slows down inference.

If the LLM needs to process more information than its KV cache allows, there are two main strategies for achieving this: either compressing the KV cache or using external long-term memory. The former includes KV quantization (Liu et al., 2024; Hooper et al., 2024; Shutova et al., 2025), pruning (Zhang et al., 2023b; Xiao et al., 2023), offloading techniques (Lee et al., 2024) and other techniques that reduce the overhead from having a large KV cache. The latter bypasses the need for large KV cache by allowing the LLM to offload its knowledge to an external database and retrieve it later.

Memory-Augmented LLMs. Even before the LLMs, NLP researchers have allowed their models to retrieve relevant knowledge from a corpora to give it access to additional knowledge and improve factuality (Guu et al., 2020; Karpukhin et al., 2020; Lewis et al., 2020). However, not all problems can be solved with a static database: for instance, when dealing with user preferences or working memory, the model must be able to update its knowledge to maintain relevant information (Wu et al., 2022). To address this, Packer et al. (2023) propose MemGPT — an external memory system that equips LLM agents with tools to create and update memory entries for later retrieval.

Since then, multiple lines of work propose extensions to this idea using knowledge graphs (Rasmussen et al., 2025; Chhikara et al., 2025; Wang et al., 2024a), hierarchy (Sun & Zeng, 2025; Kirmayr et al., 2025; Zhang et al., 2026a) insights from cognitive science (Nan et al., 2025), note taking (Xu et al., 2025; Tekparmak & Ömer Kaya, 2025), and others (Gao et al., 2025; Fang et al., 2025; MemMachine Contributors, 2025). Aside from storing user preferences, agentic memory is also used for knowledge updates (Wang et al., 2024b; 2025b), iterative “evolutionary” problem solving (Agrawal et al., 2025; Zhang et al., 2026b), programming (Gandhi et al., 2025; Shen & Joung, 2025), embodied and virtual agents (Glocker et al., 2025; Lei et al., 2025; Wang et al., 2024c; Qi et al., 2025). Recent works propose even more complex memory frameworks and operating systems (Li et al., 2025; Kang et al., 2025) or fine-tuning the LLM to a specific memory type (Modarressi et al., 2024; Yan et al., 2026).

Benchmarking Long-term Memory frameworks is an inherently difficult task. Two of the most popular benchmarks for memory agents are LOCOMO (Maharana et al., 2024) and LongMemEval (Wu et al., 2024), focus on a chat assistant use case where the agent is expected to recall individual facts, do multi-hop look-ups and update its knowledge over time as the context changes. Other benchmarking scenarios are based on interactive conversations with user feedback (Kim et al., 2024; Wan & Ma, 2025; Ai et al., 2025) or convert existing long-context datasets (Hu et al., 2025). Other more specialized benchmarks target test-time learning (Wei et al., 2025), personalization and others (Kim et al., 2025; Jiang et al., 2025; Jia et al., 2025). However, recent works have found that many of these evaluation tasks do not use complex memory hierarchies (Zhou & Han, 2025; Ai et al., 2025; Rasmussen et al., 2025). Notably, Zhou & Han (2025) introduces EMem and Emem-G: simple retrieval baselines that outperform more complex memory structures on both LOCOMO and LongMemEval. We hypothesize that these benchmarks do not test the more advanced components of agentic memory, making it harder to analyze their effectiveness.

3 STRUCTMEMEVAL

In this work, we create a benchmark for evaluating one specific capability of memory agents: how well can they structure their long-term memory for a given problem. The main difficulty for such benchmark is that we cannot rely on a specific internal memory architecture. As we discussed in Section 2, modern memory frameworks use varied internal representations from look-up dictionaries, Zettelkasten notes, to graphs and temporal databases. Given the same task, Mem0^g and Zep will map the desired knowledge structure into a graph database, whereas A-Mem and MemAgent would use interlinked notes. To compare different memory frameworks, we need our task suite to be implementation-agnostic. To that end, we only evaluate the final answers, not the internal structure, but we design our problems in such a way that solving them requires the agent to organize its memory.

¹E.g. storing 1M tokens for Qwen2.5-7B-Instruct-1M in bfloat16 takes up over 180GiB GPU memory.

Another important consideration is that our tasks need to benchmark the memory structure specifically. For instance, if we were to give the LLM an extremely difficult programming task with limited context window, the agent would have to use long-term memory. However, it would also require programming and design capabilities, which would contaminate our results. If an agent failed to solve the problem, it would be unclear if the error was caused by bad memory organization or simple coding error. To decouple our benchmark from other capabilities, we deliberately select tasks that are simple given correct memory organization but nearly impossible without it. For convenience, we organize our benchmark around specific memory organization patterns that humans have already used: trees, ledgers, to-do lists, indexes and others. For each pattern, we gather problems where a human expert would follow that structure in their notes. We discuss each structure type below.

Tree-structured problems require the agent to maintain hierarchical knowledge such as taxonomies, family trees, corporate hierarchies or codebases. To test this ability, we design problems based on family trees and corporate hierarchies. The memory agent is given a sequence of messages that encode relations (“A is B’s stepdaughter”) and is expected to maintain the full graph. Some of the relations are implied (e.g. if “C is B’s wife”, then the last example also implies that “C is A’s parent”). Notably, we found that such indirect queries often confuse purely retrieval-based LLMs.

State tracking problems are tasks where one or multiple entities can change their state over time. A practical example is when someone moves cities, they are then no longer neighbors with those left behind. A similar state transition can happen with tasks on KANBAN boards or functions during codebase refactoring. In our evaluation, we gather tasks where individual messages only matter in context of their state and keeping track of state transition is important for answering the question.

Counting problems are tasks that involve maintaining and reconciling totals. For example, an accountant may use offsetting or netting diagrams to determine final settlement amounts (i.e., which party owes how much), alongside other task-specific organization practices (Kieso et al., 2020). Similarly, a personal recommender assistant would keep track of their user’s experiences to determine which activity it should recommend, a scientist would keep statistics of their observations. We gather tasks where the agent observes a history of transactions (e.g. “A supplied \$X for B”) between multiple parties, and must then compute the final settlement after netting (i.e. canceling circular debts).

Data curation. We gather scenarios using both human annotators and LLM augmentation with human verifiers. Every scenario consists of a conversation history and a set of evaluation questions asked at different conversation depths: testing if two people are related for trees, evaluating the final debt settlement amount, etc. Using synthetic scenarios allows us to release tasks themed after sensitive user and business applications without data privacy risks (Feretzakis et al., 2024; Goyal & Mahmoud, 2024). We evaluate accuracy using exact match when possible (e.g. yes/no or numeric answers) and LLM-as-a-judge (Zheng et al., 2023) for questions without a clear answer format. Overall, we gathered 73 unique evaluation scenarios (conversations) with 544 evaluation questions. We include additional details on data gathering and curation for every task type in Appendix A. In future, we plan to extend this with additional problems: ordered lists for to-do queues and finding the K best candidates, assignment maps for calendars and resource allocations, and others.

Memory organization hints. For each scenario, we provide an optional “memory organization hint”: an informal text prompt that explains how a human would organize their knowledge for the task at hand. We evaluate StructMemEval in two modes: with and without hints. Our *main setup is evaluating without hints*. Instead, we use structure hints to diagnose error modes: if an agent fails to solve a given problem as is, but then solves it reliably with the hint, then its original mistake stemmed from poor knowledge structure. If, however, that agent still fails even with “hints”, then the error is not due to wrong memory organization strategy, but in poor execution: failing to maintain the chosen memory structure or failing to properly utilize it when answering the test questions.

In our analysis, we evaluate the following memory frameworks: 1) retrieval-augmented LLM based on OpenAI embeddings² (OpenAI, 2023) with different retrieval budgets, 2) markdown-based memory from Mem-agent (Tekparmak & Ömer Kaya, 2025) and 3) Mem0 agentic memory (Chhikara et al., 2025). Appendix B contains additional details and configurations for each memory framework. We run our main experiments using gemini-2.5-pro and gemini-3-pro (Comanici et al., 2025) backbone LLM, depending on the experiment. We evaluate other LLM backbones in Appendix C. We evaluate memory frameworks with default hyperparameters, leaving room for future tuning.

²We use text-embedding-3-large model and the retrieval baseline code from Chhikara et al. (2025)

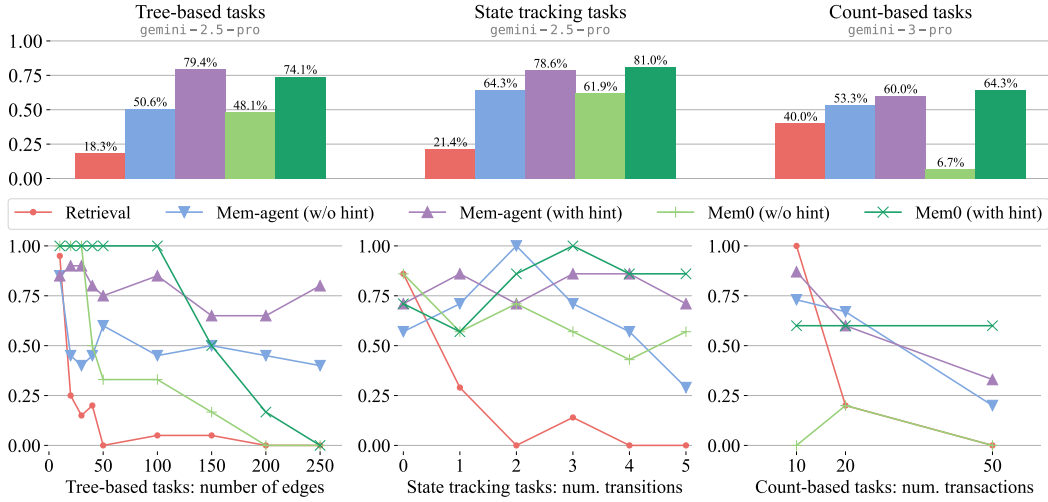


Figure 1: **Top:** accuracies across task types for retrieval-augmented LLM, Mem-agent, and Mem0. **Bottom:** detailed accuracies per difficulty level (see X axis label) using the same colors as above.

4 EXPERIMENTS

We evaluate our task suite with both simple retrieval and memory agents. Figure 1 summarizes our results for gemini 2.5 & 3 pro LLM and text-embedding-3-large retrieval. We use gemini-3 for count-based tasks because gemini 2.5 has too many hallucinations (see Table 2). The retrieval system uses top-15/10/5 search for tree-based, count-based and state tracking respectively. We report additional models (gemini-3-flash and gpt-4o-mini) and top- k configurations in Appendix C.

As expected, simple retrieval-augmented agents can solve small-scale problems, but fall off when the task exceeds their retrieval budget. For state tracking problems, retrieval often takes messages out of context. For instance, in a scenario where 1) the user interacted with their neighbors, then 2) moved to a different location and interacted with new neighbors there, before 3) returning to their original place, retrieval often prioritizes mentions of the term “neighbor” from stage (2) even though these are no longer the user’s neighbors due to stage (3). In contrast, both Mem-agent and Mem0 score much better across all three problem types *when given a memory organization hint*. Without the hint, both agents still outperform retrieval, but their answers are less reliable. This seems counterintuitive because LLMs are known to solve complex programming tasks using similar data structures: trees, heaps, state machines (Chen et al., 2021; Jain et al., 2025). We hypothesize that LLMs were simply not trained to apply algorithmic knowledge to their own memory and may improve with fine-tuning.

Curiously, the gap between memory agents with and without hints is far larger than the difference between Mem0 and mem-agent. When evaluating other models in Appendix C, we found that this gap persists. Newer model generations are better at following hints, but they still often fail to properly organize their memory without hints. After analyzing LLM behavior that led to incorrect answers, we found two main failure modes: either (i) the LLM does not organize its memory, especially when not hinted or (ii) the LLM hallucinates spurious memories, which happens rarely in normal use, but more often when the LLM performs hundreds of consecutive memory updates. See Appendix D for details.

5 DISCUSSION & FUTURE WORK

In this work, we study how well modern memory agents can organize their knowledge. We introduce StructMemEval — a benchmark suite evaluating an agent’s ability to organize memory into structures that facilitate specific tasks. We evaluate on three distinct problem subsets and find that memory agents can significantly outperform simple retrieval if they adhere to an appropriate memory structure. However, we also observe that modern LLMs do not always structure their memory correctly unless explicitly prompted to do so. This highlights two critical directions for future research: **training (or prompting) backbone LLMs to better structure knowledge within existing frameworks**, and **designing memory systems that intrinsically facilitate this capability**. In future work, we aim to **extend StructMemEval with additional memory patterns**—such as ordered (to-do) lists, DAGs, assignment maps, indexes, as well as tasks that require managing multiple structures simultaneously.

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A DETAILS ON DATA COLLECTION

In this section, we describe the data gathering protocol for each task subset in StructMemEval. Our approach consists of the following stages: i) manually create initial scenarios to test their feasibility for different memory types, ii) use LLM augmentations to create multiple similar scenarios and additional evaluation questions iii) manually verify the resulting scenarios to fix hallucinations and make sure that the reference answers are valid. However, there are slight variations in methodology depending on the task at hand. We also use 3 different LLMs for augmentation to reduce the risk of style contamination implicitly favoring one model over another (Panickssery et al., 2024).

Accounting problems: we start by manually specifying examples of transactions (e.g. "Alice gave Bob \$N for ..."), then use deepseek/deepseek-v3.1-terminus to generate more transactions similar to the manually created ones. We also introduce unrelated messages, creating scenarios with around 10%, 30% and 50% unrelated messages (5 unique scenarios each, for a total of 15). For these tasks, the only task for the model is to determine the final settlement between parties (with netting) and test if it matches the correct one. We ask for the total settlement amounts multiple times after certain numbers of transactions (conversation depth) to create tasks of different complexity. To determine the reference settlement, we run the same LLM in agentic setup, giving it tools to compute settlement from individual transactions, then verify and correct answers manually. This is because, without tools and verification, even full-context reference LLM sometimes hallucinates transactions.

Recommendation problems: for this subset, we consider tasks where the LLM assistant interacts with a user who watches movies, reads books or attends other activities over a prolonged period of time, after which the agent must decide what activities the user tried more and which ones did they like more on average, expecting an exact numeric answer. This task is difficult for retrieval-augmented LLMs since they can't calculate statistics for top-k search results if the sample size is larger than k . To gather these tasks, we follow a similar pattern to how we did accounting problems, except that we use qwen/qwen3-max to generate dialogues, then verify with deepseek/deepseek-v3.1-terminus.

Tree-based problems: for relations and family tree tasks, we first generate a reference graph corresponding to a family tree where each node corresponds to a person. For simplicity, we ensure that all persons have unique names to avoid confusion. For each graph, we pick pairs of nodes for which there is a unique shortest path between them that has 10 hops (11 nodes visited). Then, we create subsets of graphs containing 10, 20, 30, 40, 50, 100, 150, 200 and 250 links (encoded as messages). We ensure that each subset includes the shortest path between the pair of nodes that we picked previously. After the agent processes these messages, we ask it to find the shortest path. For augmentation, we use the same DeepSeek model as in the accounting subset, but we allow tool use.

State tracking problems: finally, we gather tasks that require the agent to track state changes. Overall, we adopt a similar pipeline to our accounting tasks, but using anthropic/claude-opus-4.5 (through Claude Code) for data augmentation. For this evaluation, we start with control tasks with no 0 changes ("static") and consider progressively more complex transitions up to 5 state changes. Tasks with 1 transition have one state change that affects the final outcome, such as with the "I owe \$X to my neighbor; I moved to a different city Y, do I owe something to my neighbor?", and subsequent difficulty levels feature multiple such transitions, all of which are necessary to solve the problem. Unlike accounting tasks, we do not reuse conversations between difficulty levels and instead create different unique conversations for each number of transitions. Table 1 summarizes data collection statistics. Curiously, note that the conversations with 2 state transitions are shorter than 1 transition tasks in terms of the average number of messages, but models still find them more difficult.

Table 1: Data gathering statistics for state tracking problems.

Difficulty	Files	Sessions	Total Msgs	User Msgs	Avg. msgs/session	Avg. msgs/file
static	7	15	150	75	10.0	10.7
1tr	7	35	219	110	6.3	15.7
2tr	7	36	188	94	5.2	13.4
3tr	7	49	238	119	4.9	17.0
4tr	7	59	296	148	5.0	21.1
5tr	7	73	362	181	5.0	25.9
TOTAL	42	267	1453	727	5.4	17.3

B DETAILS ON MEMORY FRAMEWORKS

In Section 4, we evaluate the following LLM configurations:

- **Retrieval-augmented LLM:** we considered several implementations for purely retrieval-augmented LLMs and eventually found that Mem0’s built-in retrieval module over OpenAI’s text-embedding-3-large embeddings works well on standard benchmarks. To evaluate this configuration, we use Mem0 codebase³ version 1.0.2, run the built-in search module, then feed the search results into the main LLM to compose the final answer. We search as follows:

```
import mem0
m = mem0.memory.main.Memory(
    mem0.configs.base.MemoryConfig(
        llm=mem0.llms.configs.LlmConfig(provider=..., model=..., api_key=...), # main LLM
        embedder=mem0.embeddings.configs.EmbedderConfig(
            provider="openai", config={"model": "text-embedding-3-large", "api_key":...}),
        vector_db=mem0.vector_stores.configs.VectorStoreConfig(
            provider="qdrant", config={"collection_name": ..., "path": ..., # random uid
            "embedding_model_dims": 3072},
        )
    )
)
m.reset()
m.add(FEED_CONVERSATION_HERE, user_id="same_user_everywhere", infer=False)
results = m.search(question, user_id="same_user_everywhere", limit=TOP_K_LIMIT_HERE)
```

- **Mem-agent memory:** for a simple abstract memory framework, we use the markdown-based memory from Tekparmak & Ömer Kaya (2025) using their official codebase⁴. This memory framework allows the agent to create, modify and retrieve any number of markdown files locally. While the original study focuses on teaching small LLMs (Qwen3-4B) to use this memory, we found that google/gemini-2.5-pro can already use the memory well with the prompt alone: it solves simple retrieval tasks well and, when it fails on our tasks, it is because of a knowledge organization error or hallucination, not failure to use the framework itself. Our main rationale for experimenting with Mem-agent is that it is extremely easy to use: pure-python using files, no need to spin up services before use. However, as our experiments demonstrate, it can implement proper memory organization patterns when hinted to do so, which means that it is flexible enough for our tasks. When evaluating mem-agent, we instantiate `mem_agent = agent.agent.Agent(model=MAIN_LLM_HERE)` and use `mem_agent.chat(input_shard)` to both encode memory and answer questions. Crucially, we clear agent’s recent message history (`mem_agent.messages = mem_agent.messages[:1]`) after each conversation message. That way, we can be certain that the agent answers the question from memory, not by directly accessing conversation history. This corresponds to a use case where the messages are spread over multiple conversations over a long time-frame and are not neatly packed into one session. **When evaluating with hints**, we additionally modify the agent’s system prompt (first message) and append the task-specific hint after the default mem-agent prompt.
- **Mem0 agentic memory:** Finally, we evaluate the full agentic Mem0 memory. For this evaluation, we use the same search module as in our earlier retrieval-based evaluation, but we allow the LLM to decide which memories to store and how to formulate questions for retrieval, allowing the LLM to create, manage and retrieve memories. We use the official codebase (Chhikara et al., 2025) with default parameters. Note that both Mem0 and other frameworks will likely work better after tuning.

On the (lack of) hyperparameter tuning. In this initial evaluation, we deliberately evaluate memory as is with minimal tuning. We urge the readers not to draw conclusions such as “Memory A is better than memory B because of this experiment.” There are two main reasons for this:

- **Framework configuraiton:** Popular long-term memory frameworks typically have a large number of “knobs” to tune to optimize the framework for the given task. We avoid this in our initial analysis.
- **Model dependency:** we run our experiments with gemini-2.5-pro with some additional experiments with gemini-3-pro and gpt-4o-mini in Appendix C. Other models can score differently.

³<https://github.com/mem0ai/mem0>

⁴<https://github.com/firstbatchxyz/mem-agent>

C ADDITIONAL MODEL EVALUATION

C.1 COUNT-BASED TASKS

Table 2: Early evaluation for gemini-2.5-pro and different retrieval budgets on count-based tasks for 10% and 30% noise (we omit 50% as non-informative). Agentic memory fares poorly across all budgets due to hallucinations. Gemini 3 pro scores significantly higher (see Table 3 below).

Noise level	10%			30%			Average		
Transactions	10	20	50	10	20	50	10	20	50
Retrieval (top-5)	0	0	0	0	0	0	0	0	0
Retrieval (top-10)	0.6	0	0	0.6	0	0	0.6	0	0
Retrieval (top-20)	1.0	0.4	0	0.6	0.6	0	0.8	0.5	0
mem-agent (w/o hint)	0	0	0	0	0	0	0	0	0
mem-agent (with hint)	0.2	0	0	0.2	0.2	0	0.2	0.1	0

Table 3: Detailed evaluation for gemini-3-pro and different retrieval budgets on count-based tasks.

Noise level	10%			30%			50%			Average		
Transactions	10	20	50	10	20	50	10	20	50	10	20	50
Retrieval (top-5)	0	0	0	0	0	0	0.6	0	0	0.2	0	0
Retrieval (top-10)	1	0	0	1	0	0	1	0.6	0	1	0.2	0
Retrieval (top-20)	1	0.6	0	1	0.6	0	1	1	0	1	0.73	0
mem-agent (w/o hint)	0.8	0.6	0.2	0.8	0.6	0	0.6	0.8	0.4	0.73	0.67	0.20
mem-agent (with hint)	0.8	0.8	0.4	1	0.6	0.2	0.8	0.4	0.4	0.67	0.47	0.27

C.2 STATE TRACKING TASKS

Table 4: Accuracy on state tracking tasks by model, **tr** is the number of relevant state transitions. Per-transition results are noisy due to small sample. Avg. is the mean accuracy over all scenarios.

Model	Memory System	0tr.	1tr.	2tr.	3tr.	4tr.	5tr.	Avg.
gpt-4o-mini	retrieval (top-5)	86%	43%	0%	14%	0%	0%	24%
	retrieval (top-20)	86%	71%	43%	29%	14%	0%	40%
	mem-agent (w/o hint)	71%	43%	86%	29%	71%	43%	57%
	mem-agent (with hint)	86%	71%	86%	43%	57%	100%	74%
	Mem0 agent (w/o hint)	71%	57%	29%	14%	0%	0%	29%
	Mem0 agent (with hint)	43%	57%	71%	71%	71%	43%	60%
gemini-2.5-pro	retrieval (top-5)	86%	29%	0%	14%	0%	0%	21%
	retrieval (top-20)	71%	71%	14%	0%	0%	0%	26%
	mem-agent (no hint)	57%	71%	100%	71%	57%	29%	64%
	mem-agent (with hint)	71%	86%	71%	86%	86%	71%	79%
	Mem0 agent (w/o hint)	86%	57%	71%	57%	43%	57%	62%
	Mem0 agent (with hint)	71%	57%	86%	100%	86%	86%	81%
gemini-3-flash	Mem0 agent (hint)	71%	100%	100%	100%	100%	100%	95%

D ERROR ANALYSIS

We identified three categories of errors in the agentic memory files for the most demanding high precision benchmark subset – accounting problems: 1) the omission of transactions, which was the most common; 2) the duplication of existing transactions; and 3) most significantly, the hallucination of transactions, as illustrated in Table 5.

Table 5: Example memory entries for mem-agent generated for accounting problem 11 with gemini-2.5-pro. There is a transaction with the note “sandwich” – but no line in this conversation the word “sandwich”, and there is no real transaction with its listed payer/amount. Moreover, the first and last rows are duplicate entries for the same transaction.

Payer	For Whom	Amount	Currency	Notes
<i>Bob</i>	<i>All</i>	<i>57</i>	<i>EUR</i>	<i>lunch</i>
Bob	All	51	EUR	snacks
Alice	Alice, Bob	84	EUR	dinner
Alice	All	120	EUR	train tickets
Charlie	Bob, Charlie	60	EUR	groceries
Alice	All	45	EUR	Museum entry
Charlie	All	75	EUR	Lunch
Alice	Alice, Charlie	-15	EUR	refund on museum tickets
Bob	All	30	EUR	taxi
Alice	All	90	EUR	Dinner
Bob	Alice, Bob	36	EUR	breakfast
Alice	Bob	15	EUR	sandwich
<i>Bob</i>	<i>All</i>	<i>57</i>	<i>EUR</i>	<i>lunch</i>